CCCA Cell-State Analysis — Bischoff 2021 (LUAD)

Lung adenocarcinoma · 10x · 13,382 malignant cells (10 tumour samples, ≥44-cell threshold)

Verdict: embryonic pattern does NOT reproduce — malignant structure is two disconnected EMT/differentiation lobes, not one continuous manifold; no trajectory fit (Gate 1 NO-GO).

Gate ledger

NO-GO **Gate 1 (mixed + continuous?)**

MIXED ✓ diff-sample kNN frac 0.02→0.47 (ceiling 0.81); both lobes internally sample-mixed, tumours span both.
CONTINUOUS ✗ two islands: PAGA between-lobe 0.01 vs within 0.55; 0.18% of edges & 1.9% of cells bridge; bimodal diffusion axis (peak/valley 387×). Invariant across θ∈{0,2,5,10}, n_pcs∈{12,16,20} → not under-integration.

NA **Gate 2 (hub → two arms, Crest at tips?)**

Not evaluated — Gate 1 NO-GO. No manifold/signature step produced (spec §6).

NA **Gate 3 (Gast↔pSom cross-repression?)**

Not evaluated — branching-only, and no trajectory exists.

Lobe-identity fork (Kürten test): is either lobe alone a valid scope target?

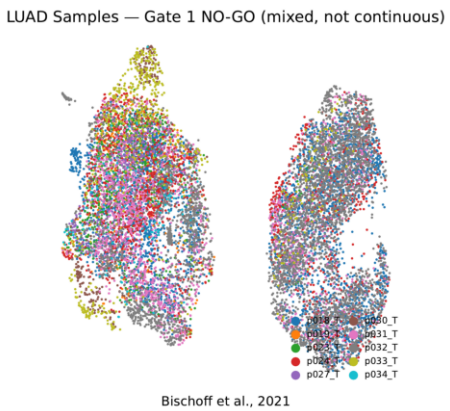
Signatures near union-control baseline everywhere: Gast −0.04, pSom +0.06, Crest −0.03 (std ≈0.034) — programs not differentially active in this malignant compartment.
Alveolar lobe shows the Kürten sign pattern (Gast~pSom +0.22, both oppose Crest −0.56/−0.72) but on baseline magnitude, and it is the DIFFERENTIATED pole — biologically backwards for a recapitulation claim.
EMT/dediff lobe is incoherent (Gast~pSom −0.32).
→ No lobe reproduces the reference geography with real signal → no valid scope target (unlike Kürten's Embryonic lobe).

Key parameter choices

Min-cell threshold	≥44 (geom-mean of 127→15 gap, 8.5×) → 10/19 samples
Over-dispersed genes	1,976 (scFates GAM)
n_pcs	16 (PCs > 0.5% variance; PC1 = 12.4%)
Integration	Harmony on sample, θ=2 (mixing plateaus by θ=2)
Embedding	neighbours k=15, UMAP spread=1.3, Leiden res=0.5 (12 clusters)
Lobe split	diffusion-axis valley: Alveolar 6,100 / EMT-dediff 7,282
Signatures	score_genes vs union pool (Gast 80 / pSom 95 / Crest 103 genes)
EMT-TFs shown	SOX4 83%, SOX9 44%, TEAD1 44%, FOXC1 33% (≥3% floor; VIM excluded)

Step figures

01 · samples.pdf



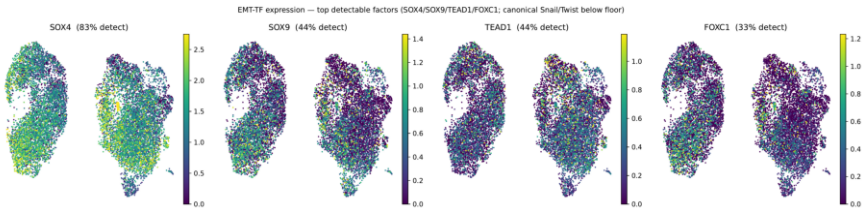
01_preprocessing → samples.pdf

Two lobes, each internally sample-mixed (salt-and-pepper) — not patient islands. The empty waist between them is the discontinuity that fails Gate 1.

02_expression → expression.pdf

Descriptive EMT-TF overlay (no trajectory to polarise along): SOX4 graded high on the EMT/dediff lobe; SOX9/TEAD1/FOXC1 patchier.

02 · expression.pdf



Gate 1 NO-GO · deliverables: 01_preprocessing + 02_expression (spec §6) · fully-autonomous run, attempt 2